

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.

16 Port DO Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates '**resettable Fuses**' to safeguard against sudden High Voltage spikes. the board also has '**reverse polarity**' protection both for input and power connector.

the communication port also has signal isolation and additional jumper for 'end of line' 120 ohms termination resistance which may or may not be used.

Specifications

General –

I/O Connectors	16 Pin 5.08 mm pitch pluggable screw Terminal Block.
Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 - 24 VDC or 24 V AC/DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



Relay Output –

Channels	16
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10 ⁵
Mechanical Life	1 x 10 ⁷
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

Additional Features: -

All inputs and communication ports isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

DIP SETTING – For Manual Switch Setup**Configuration Settings: -**

Communication Speed	9600 – 115200 (DIP SW selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configurable with DIP Switch
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15
Function code D0	0x01 Read discrete Input

Note: -

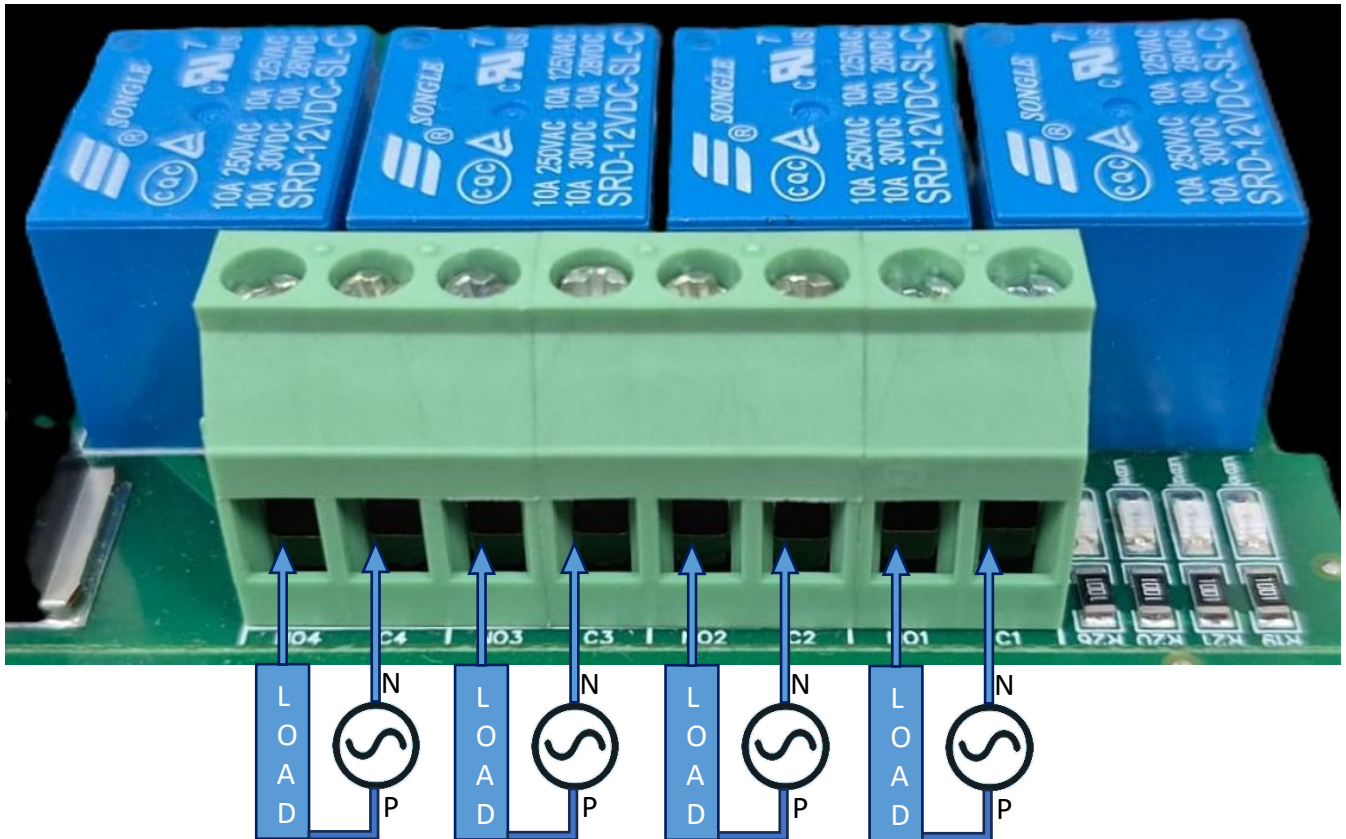
For MODBUS communications, **a shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	DO 1	00001	0X05,0X0F	RS485	1 Bit Boolean
2	DO 2	00002	0X05,0X0F	RS485	1 Bit Boolean
3	DO 3	00003	0X05,0X0F	RS485	1 Bit Boolean
4	DO 4	00004	0X05,0X0F	RS485	1 Bit Boolean
5	DO 5	00005	0X05,0X0F	RS485	1 Bit Boolean
6	DO 6	00005	0X05,0X0F	RS485	1 Bit Boolean
7	DO 7	00007	0X05,0X0F	RS485	1 Bit Boolean
8	DO 8	00008	0X05,0X0F	RS485	1 Bit Boolean
9	DO 9	00009	0X05,0X0F	RS485	1 Bit Boolean
10	DO 10	00010	0X05,0X0F	RS485	1 Bit Boolean
11	DO 11	00011	0X05,0X0F	RS485	1 Bit Boolean
12	DO 12	00012	0X05,0X0F	RS485	1 Bit Boolean
13	DO 13	00013	0X05,0X0F	RS485	1 Bit Boolean
14	DO 14	000014	0X05,0X0F	RS485	1 Bit Boolean
15	DO 15	00015	0X05,0X0F	RS485	1 Bit Boolean
16	DO 16	00016	0X05,0X0F	RS485	1 Bit Boolean

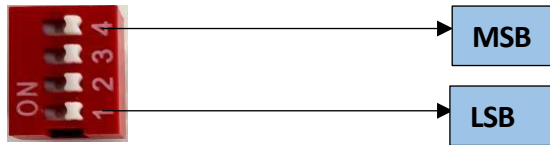
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C** and **NO** remains open, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The load is connected between terminal **NO** and **Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The load is connected between terminal **NO** and **Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

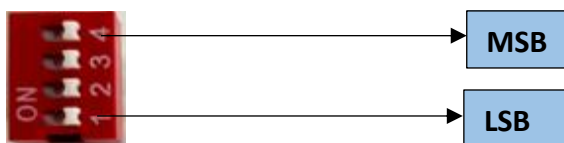
BAUD RATE DESCRIPTION:



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When u change the Baud rate in the Module power 'ON' condition, pls press the reset button to get Change to affect

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
19200	ON	OFF	OFF	OFF
38400	OFF	ON	OFF	OFF
57600	OFF	OFF	ON	OFF
115200	OFF	OFF	OFF	ON

SLAVE ID DESCRIPTION



For Slave ID Selection SW is used to Set The SLAVE ID .

For Slave ID DIP Switch **LSB is "1"** follow through **"4" is MSB**.

Slave ID Confirmed through below Device ID table .

IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So "1 0 0 0" will be selected as Slave ID 1.

Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 0 1 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	0 0 1 1	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 1 0 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	0 1 0 1	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	0 1 1 1	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	ON(1)	OFF(0)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

SOFT SETTING – For Modbus / Software Setup

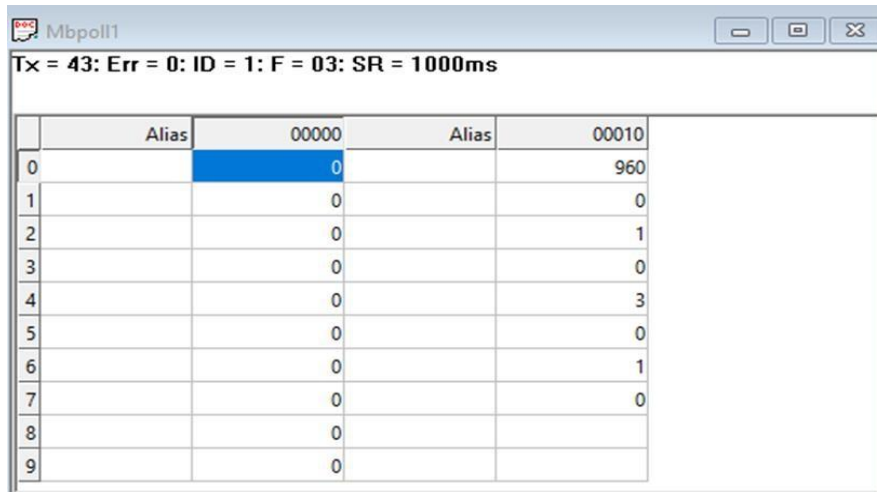
Configuration Settings: -

Communication Speed	9600 – 115200 (DIP SW selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Soft settings (0-255).
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15.
Function code DO	0x01 Read Discrete Input

Note:

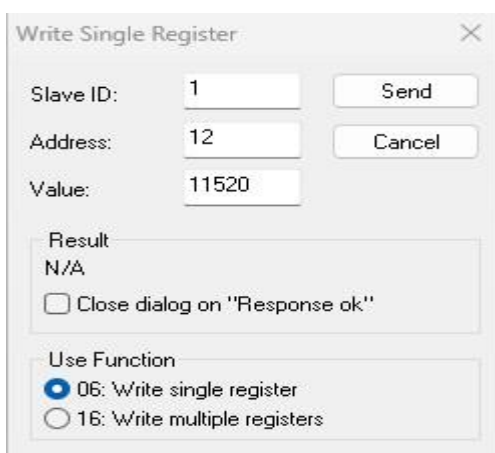
1. In register 40015- you will be able to see the current parity and you can change it in register 40016
2. In register 40017- you will be able to see the current stop bit and you can change it in register 40018
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc. Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system.

BAUD RATE and SLAVE ID DESCRIPTION:



	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

- As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id you have to enter a value on the 40012, resistor for the baud rate and 40014 for the slave id as shown in the image below.



Write Single Register

Slave ID: 1

Address: 12

Value: 11520

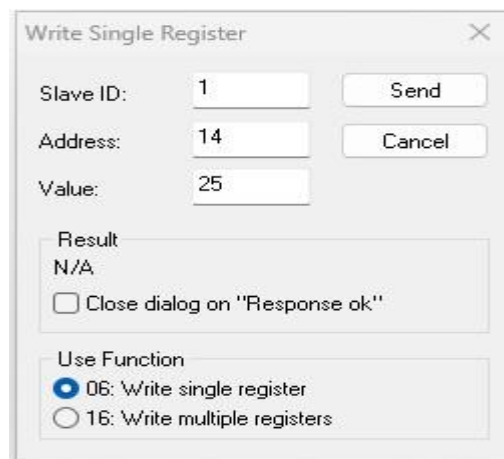
Result
N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers



Write Single Register

Slave ID: 1

Address: 14

Value: 25

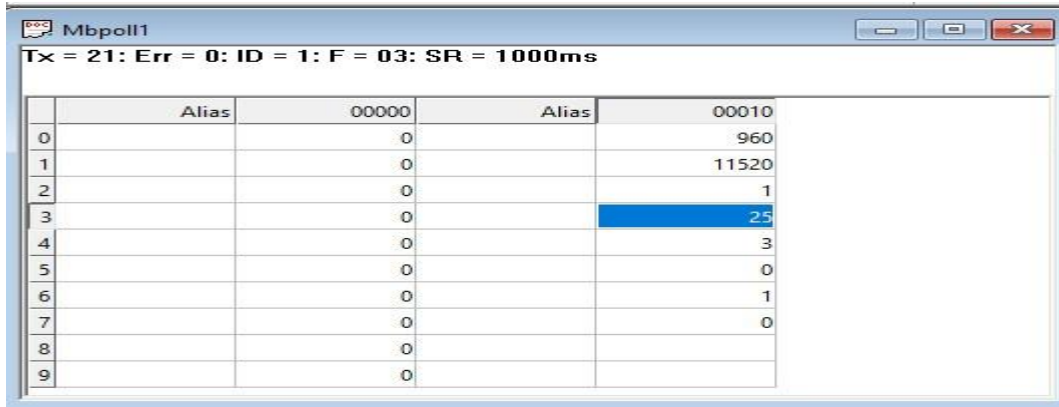
Result
N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers

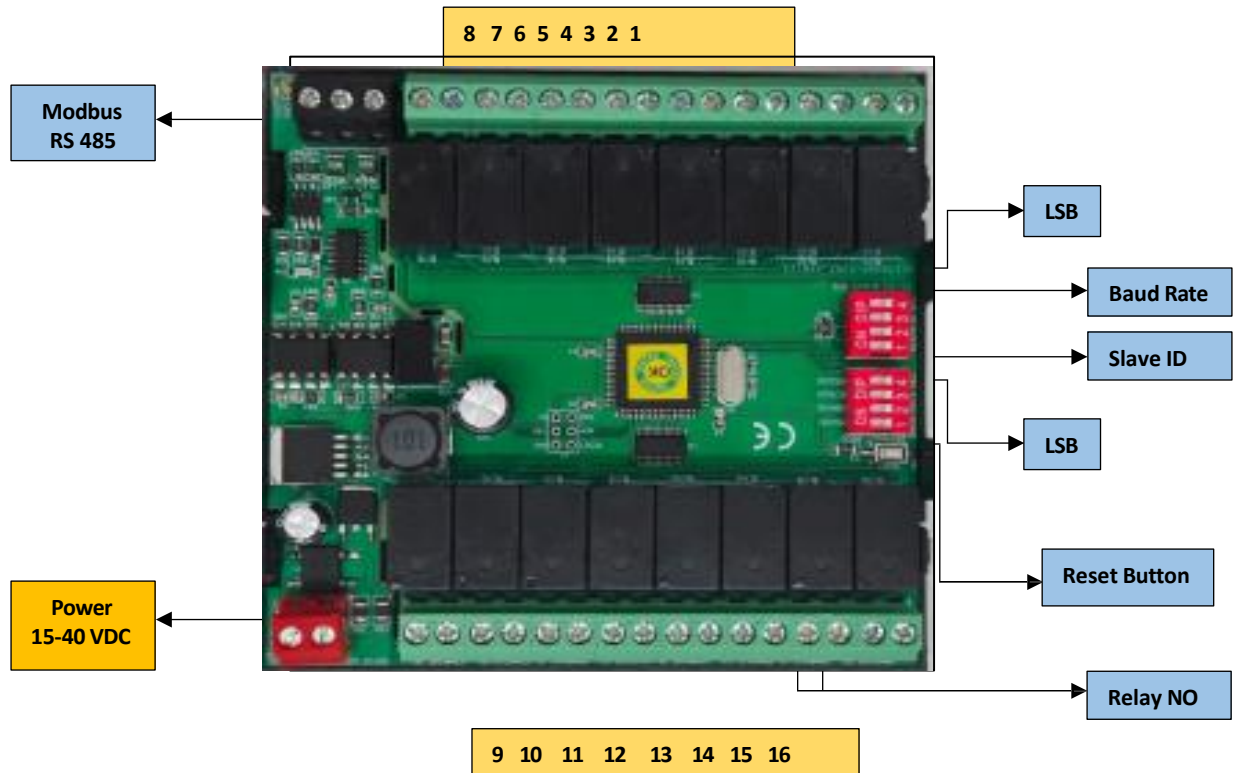


	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above I set 115200 baud rate and slave ID 25.

Note:

- When entering baud rate values such as 9600 / 19200 / 38400 / 57600 / 115200, enter them as 960 / 1920 / 3840 / 5760 / 11520, respectively.
- After entering the values, press the reset button or reboot the system.
- Once restarted, you will see the updated values:
- 40011 – 11520 → Baud Rate (115200)
- 40013 – 25 → Slave ID (25)
- You can also change Parity Bit and Stop Bit of the device in the same way.
- Jumper Setting (Hardware Method):
- If you place the jumper, the device will load its default communication settings (default baud rate (9600) and slave ID (1)).
- This is useful when you have many devices with different settings and want a quick way to restore defaults without software changes.
- Relation between Jumper and Reset Button:
The jumper enables the default mode, and the reset button activates it.
- Simply insert the jumper → press reset → the device restarts with factory default settings automatically



For More Details: <http://www.youtube.com/@wittelb>

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